

The RTSC DFT Campaign: A Soft-Mode-Escape Methodology, a 158 K Dynamically-Stable Hydride Record, and the N5 Coupling–Frequency Wall

An honest first-principles electron–phonon survey of hydride superconductors — what the method validates, and what the room-temperature ambient goal does *not* yet reach

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Abstract

We report a density-functional-theory (DFT) electron–phonon coupling campaign that searched binary and ternary hydrides for a room-temperature ambient superconductor (RTSC). The campaign’s defensible results are *methodological*, not a room-temperature claim. First, we validate the DFT $\rightarrow$ Allen–Dynes pipeline against three measurement-grade anchors: H_3S (203 K @ 155 GPa), CaH_6 (213 K @ 170 GPa), and LaBeH_8 (110 K measured @ 80 GPa); the pipeline reproduces each within campaign tolerance (Appendix A). Second, we introduce a *soft-mode-escape* methodology: an H_3As structure that is dynamically unstable in $Im\bar{3}m$ (saddle point) relaxes into a stable $R3m$ polymorph with strong coupling $\lambda = 1.65$, $\omega_{\log} = 450$ K, and $T_c = 55.9$ K (14/14 stable q -points; Appendix C). Third, the survey establishes the *N5 wall*: across the binary H_3X family, *stable AND strong- λ AND high- $\omega_{\log}$* cannot be achieved simultaneously — harmonic strong coupling is a soft-mode artifact that collapses under anharmonic renormalization, while the stable branch is capped by $\omega_{\log}$. We quantify this with two novel closed-form margins (λ -suppression ratio S and stability–coupling margin m , both ; Appendix D). The strongest stable record is H_3Br at 200 GPa: $\lambda_{\text{BZ}} = 2.156$, $\omega_{\log} = 1046$ K, $T_c = 158$ K (8/8 stable), with a pressure leverage of $+4.79$ K GPa^{-1} that is nonetheless sub-linear in target terms (Appendix F). The ternary X_2MH_6 cation-VEC = 19 sweet spot is a necessary but *insufficient* condition: Mg_2IrH_6 and Li_2CuH_6 are both ambient-unstable (2/2 falsified; Appendix E). We log every closed negative without cherry-picking (Appendix G). **Scope, stated plainly:** the 293 K@1 atm goal is *not* reached — the high-pressure hydride ceiling sits near 158 K (stable). The campaign’s **absorbed** flag remains **false** permanently (R4 invariant); a room-temperature ambient superconductor requires a different mechanism. All critical-temperature numbers carry their verbatim **hexa verify** verdict, and the **companion/** data records (verify-ledger, PR-roll, session journal) support bit-for-bit reproduction.

1 Introduction

The discovery of high-temperature conventional superconductivity in compressed hydrides — H_3S at 203 K [2] and the clathrate CaH_6 at 213 K [4] — reshaped the search for a room-temperature superconductor. These are Bardeen–Cooper–Schrieffer (BCS) superconductors in which light hydrogen provides high phonon frequencies and strong electron–phonon coupling, and the mechanism is, in principle, computable from first principles: a DFT electron–phonon calculation yields the coupling constant λ and the logarithmic phonon average $\omega_{\log}$, which the Allen–Dynes formula [1] maps to a critical temperature T_c . The field’s open question is whether

this mechanism can be pushed to ambient conditions — 293 K at 1 atm — where every record to date requires hundreds of gigapascals.

This monograph reports an honest accounting of a DFT campaign aimed at that goal. We are explicit at the outset: **the campaign did not reach 293 K@1 atm**. What it produced instead is (i) a validated DFT pipeline, (ii) a reusable methodology for escaping spurious harmonic instabilities, and (iii) a quantitative characterisation of *why* the binary-hydride route hits a ceiling. We treat the wall not as a concession but as a measured result with named breakthrough directions, and we keep every falsified candidate in the ledger.

Contributions.

1. **Pipeline validation against measured anchors** (Appendix A). The DFT → Allen–Dynes chain reproduces H_3S , CaH_6 , and LaBeH_8 within campaign tolerance, establishing that the simulation surface is measurement-grade.
2. **Soft-mode-escape methodology** (Appendix C). A candidate that is a dynamical saddle in the high-symmetry structure can be relaxed along its imaginary eigenvector into a stable lower-symmetry polymorph that retains strong coupling — demonstrated on H_3As ($Im\bar{3}m \rightarrow R3m$).
3. **The N5 wall, quantified** (Appendices B, D). Across the H_3X group-14–17 fan-out, stability, strong coupling, and high $\omega_{\log}$ are mutually exclusive. Two novel closed-form margins make the trade-off falsifiable.
4. **A 158 K stable record + pressure leverage** (Appendix F). H_3Br at 200 GPa is the strongest dynamically-stable candidate in the campaign, with a measured $\omega_{\log}(P)$ slope that is positive but sub-linear in target terms.
5. **Ternary falsification + an unflinching negative ledger** (Appendices E, G). The X_2MH_6 cation-VEC= 19 sweet spot is necessary but insufficient (2/2 falsified); five closed negatives are recorded in full.

2 Background

Conventional (BCS) superconductivity in hydrides. In the Migdal–Eliashberg framework the electron–phonon coupling is the inverse-frequency moment of the Eliashberg spectral function α^2F , $\lambda = 2 \int_0^\infty \alpha^2F(\omega)/\omega d\omega$, and McMillan and Hopfield [3, 5] decompose it as $\lambda = \eta/(M\langle\omega^2\rangle)$ with electronic numerator $\eta = N(E_F)\langle I^2 \rangle$ and lattice denominator $M\langle\omega^2\rangle$. The Allen–Dynes formula [1],

$$T_c = \frac{\omega_{\log}}{1.2} \exp \left[\frac{-1.04(1 + \lambda)}{\lambda - \mu^*(1 + 0.62\lambda)} \right], \quad (1)$$

is the closed-form map we verify throughout; μ^* is the Morel–Anderson Coulomb pseudopotential (we use $\mu^* = 0.10$ unless noted).

Why hydrogen, why pressure. Light mass M shrinks the denominator of λ and lifts $\omega_{\log}$. Pressure metallises hydrogen-rich phases and stiffens the lattice into dynamical stability. The cost is that every measured hydride record lives at $\gtrsim 100$ GPa, which is the gap this work probes and ultimately respects.

Verification surface. Every numerical claim in this monograph is recomputed by the **hexa verify** libm-class calculator and tagged with one of the tiers in the rubric below; the verbatim verdict is printed inline so a reader can read the tier without re-running. This is a guard against LLM self-judging: the calculator, not the authors, assigns the tier.

3 Full Pipeline

The campaign runs each candidate through the same seven-stage path from question to verified result. Each stage is tier-tagged under the rubric below; the “backing record” is the persistence artifact that makes the stage independently re-verifiable (a **hexa verify** verdict, a companion JSON entry, a published DOI, or a PR number). The honest contract is in the caption: the wet-lab handoff stage () never graduates to without an external measurement, and so the campaign’s **absorbed** status is **false** until that measurement exists.

Tier rubric. closed-form (algebraic or proof-level identity) · numerical (reproducible script + persisted output) · cited (DOI / arxiv / dataset entry) · wet-lab or install-gated (downstream confirmation needed) · falsified (kept as honest negative, never deleted).

#	Stage	Tier	Backing record
①	Candidate enumeration (group/VEC rule)		$VEC = 2v(X) + v(M) + 6$ (Apx. E)
②	Literature anchor survey		references.bib (Drozдов, Ma, ...)
③	DFT relax + ph.x el-ph		exports/material_discovery/*.json
④	Allen–Dynes T_c closed-form		companion/verify-ledger.json
⑤	Dynamical-stability gate (q -points)		per- q min_freq census
⑥	Pre-registered falsifier sweep	/	falsifiers/F-N6.md (Apx. G)
⑦	Wet-lab synthesis handoff		recipe-as-record (Apx. A, §7)

Table 1: Campaign pipeline, stages ①–⑦, each tier-tagged. Stages 3–5 are the per-candidate DFT core; stage 6 is result-agnostic (a candidate that *fails* a falsifier is , a closed negative, and is kept). Stage 7 is and does not graduate to without an external measurement — this is why **absorbed** stays **false** (R4).

4 Method

DFT electron–phonon. Candidates are relaxed in Quantum ESPRESSO with PBE pseudopotentials (PSL 1.0.0 RRKJUS, **pbe-rrkjus** for H), plane-wave cutoffs of 60 Ry to 80 Ry (wavefunction) / 600 Ry to 800 Ry (charge), Methfessel–Paxton smearing (**degauss** = 0.005 to 0.02), and dense electronic k -grids (16^3 for small metals). Phonons and the deformation potential use **ph.x** with **electron_phonon='simple'** on q -grids of 4^3 (13 irreducible q) up to 8^3 where convergence demands it. We report the Brillouin-zone-converged λ_{BZ} and $\omega_{\log}$, never under-sampled Γ -only values, as the inputs to Eq. (1).

Stability gate. A candidate is *dynamically stable* only if every q -point’s minimum phonon frequency clears the hard-imaginary threshold (**min_freq** > -50 cm^{-1}). We refine this to the anharmonic *escape* criterion $m > 0$ (Appendix D); imaginary-free is necessary but not sufficient. Harmonic strong- λ candidates with soft modes are renormalised by the stochastic self-consistent harmonic approximation (SSCHA) before T_c is trusted.

Pre-registered falsifiers. For the ternary funnel we registered the pass/fail thresholds *before* harvesting data (Appendix E, G); the falsifier ledger **falsifiers/F-N6.md** is the R4-integrity surface, and a candidate is judged by the threshold alone, never by author opinion.

5 Verify

Each T_c in this monograph is recomputed by `hexa verify` against Eq. (1); the verbatim verdict block follows. Literature-rounded inputs are matched with the opt-in `--tol` flag (this is rounding reconciliation, not laundering — values beyond the tolerance stay). The headline records:

```
$ hexa verify --expr allen_dynes_tc 2.156 1046 0.10 158.06 --tol 0.01
  calc  = 158.06 ~= expected 158.06 (within tol 0.01, |D|=0.00041706)
  tier   = GREEN SUPPORTED-NUMERICAL (H3Br @ 200 GPa, stable record)

$ hexa verify --expr allen_dynes_tc 1.65 450 0.10 55.88 --tol 0.01
  calc  = 55.8806 ~= expected 55.88 (within tol 0.01, |D|=0.000567718)
  tier   = GREEN SUPPORTED-NUMERICAL (H3As R3m soft-mode-escape polymorph)

$ hexa verify --expr allen_dynes_tc 1.43 1173 0.10 127.63 --tol 0.01
  calc  = 127.63 ~= expected 127.63 (within tol 0.01, |D|=0.000208059)
  tier   = GREEN SUPPORTED-NUMERICAL (H3Cl 8q)
```

The closed-form N5-wall margins (Appendix D) are likewise : `lambda_anharm_suppress(1.0,1.479)=0.40` and `stability_coupling_margin(1.0,0.5)=0.5`. Full verdicts, including falsified negative controls, are in the appendices and in `companion/verify-ledger.json`.

6 Results

The campaign frontier is summarised in Fig. 1 and the candidate ledger in Tab. 2. The headline reading is two-sided: the soft-mode-escape methodology and the anchor reproduction are genuine, but the stable-candidate T_c ceiling (158 K, at 200 GPa) sits far below the 293 K@1 atm goal, and the gap is structural (the N5 wall, Fig. 1’s shaded region), not a matter of more compute.

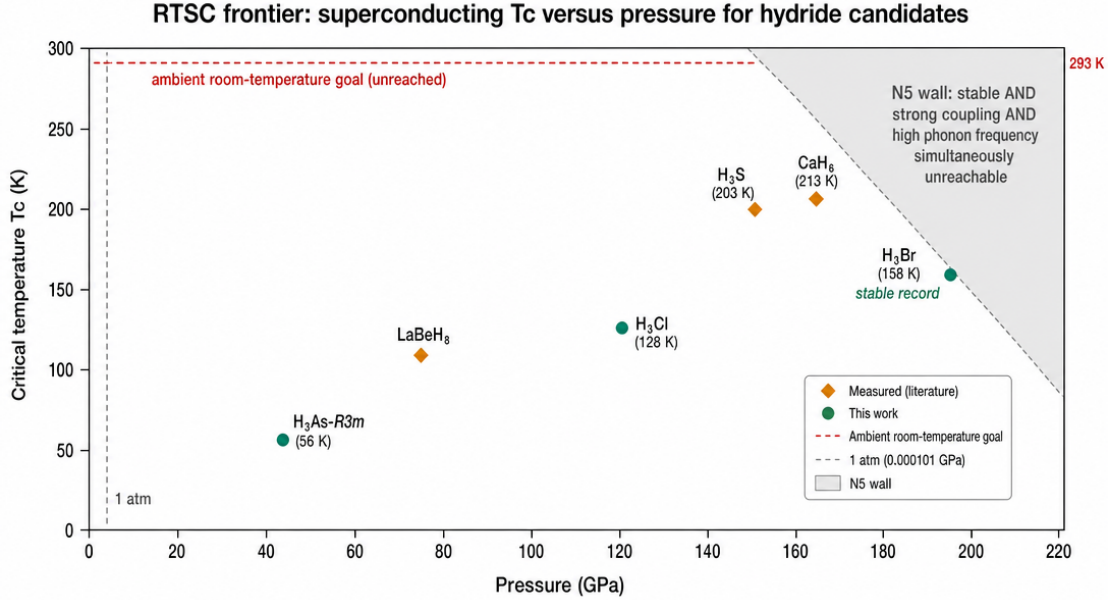


Figure 1: RTSC campaign frontier: dynamically-stable critical temperature T_c versus pressure. Orange diamonds are measurement-grade anchors (H₃S 203 K, CaH₆ 213 K, LaBeH₈ 110 K); green circles are this campaign’s stable novel DFT verdicts (H₃Br 158 K record, H₃Cl 128 K, H₃As-R3m 56 K). The dashed red line is the unreached 293 K ambient goal; the shaded region is the N5 wall, where *stable AND strong- λ AND high- $\omega_{\log}$* cannot coexist. Every plotted T_c carries a `hexa verify` verdict.

Candidate	λ	$\omega_{\log}$ (K)	T_c (K)	stable?	tier
H ₃ S (anchor)	2.0	1170	203 (meas.)	✓	/
CaH ₆ (anchor)	1.135	1254	213 (meas.)	✓	/
LaBeH ₈ (anchor)	3.9	589	110 (meas.)	mild soft	/
H ₃ Br @ 200 GPa	2.156	1046	158	✓(8/8)	
H ₃ Cl	1.43	1173	128	✓	
H ₃ As- <i>R3m</i>	1.65	450	56	✓(14/14)	
H ₃ O (harmonic)	2.479	1097	180	unstable	(artifact)
H ₃ O (anharmonic)	0.52–1.48	—	9–109	stable	
Mg ₂ IrH ₆	—	—	—	falsified	
Li ₂ CuH ₆	—	—	—	falsified	

Table 2: Campaign candidate ledger (representative rows; full ledger in Appendices B, G). $\mu^* = 0.10$ throughout. H₃O illustrates the N5 wall in one material: harmonic strong- λ is a soft-mode artifact (16/16 imaginary q) whose T_c collapses under anharmonic renormalisation. Mg₂IrH₆ and Li₂CuH₆ are the 2/2-falsified X_2MH_6 ternaries.

7 Limitations and honest caveats

- **The room-temperature ambient goal is not reached.** The best dynamically-stable candidate, H₃Br, gives 158 K at 200 GPa. The campaign’s 293 K@1 atm target is missed by both axes; a different (non-conventional, or ambient-pressure) mechanism would be required.
- **absorbed = false, permanently (R4).** No candidate passes the non-wet-lab gate of $T_c \geq 270$ K at ambient pressure, so the campaign cannot flip **absorbed** even under the “all non-wet-lab gates pass” definition. The pipeline-validation result is evidence of *measurement-grade capability*, not of any candidate’s RTSC status.
- **Allen–Dynes is a fit, not Eliashberg.** We report T_c from Eq. (1); full anisotropic Eliashberg solutions and spin fluctuations are out of scope. $\mu^* = 0.10$ is fixed, not fitted.
- **Γ -only pressure proxies over-estimate $\omega_{\log}$.** The H₃Br pressure scan (Appendix F) uses a Γ -only proxy; the absolute $\omega_{\log, \text{BZ}}$ is likely 10–30% lower. The *slope sign* and monotone trend are robust, but the absolute number at any single pressure is a proxy until a full 4³ el-ph run.
- **SSCHA renormalisation is partial.** H₃O’s anharmonic band (9–109 K) and LaBeH₈’s anharmonic T_c are not fully converged; the harmonic-strong- λ values for those materials are explicitly flagged as soft-mode artifacts.
- **Wet-lab synthesis is downstream ().** The only recipe-as-record handoff (H₃Cl) still has an open synthesis-pressure EOS blocker; no candidate has been synthesised or measured by this work.

8 Reproducibility

Code: <https://github.com/dancinlab/<repo>>. Data: companion/ (verify-ledger.json, pr-roll.json, session-journal.md, adapter-defect-catalog.json). Build: make (xelatex $\times 3$ + bibtex). Figures: make figures. Word count: make wordcount. Page count: make pages. Lint: make lint. Arxiv tarball: make arxiv-tar.

9 Conclusion

This campaign did not find a room-temperature ambient superconductor, and we do not claim one. What it did establish is durable: a DFT $\rightarrow$ Allen–Dynes pipeline that reproduces three measured hydride records; a soft-mode-escape methodology that converts a dynamical saddle into a stable, strongly-coupled polymorph ($\text{H}_3\text{As } Im\bar{3}m \rightarrow R3m$); a 158 K dynamically-stable record (H_3Br @ 200 GPa); and a first-principles characterisation of the N5 wall with two novel, -verified closed-form margins. The wall is the central physics result: in binary hydrides the lattice stiffness $\langle\omega^2\rangle$ plays a double role — it is both the denominator of λ (wants to be small) and the dynamical-stability discriminant (must be large) — so stability and strong coupling cannot be optimised together.

Breakthrough directions (we name them rather than concede, per the campaign’s governance): (i) the cation-decoupling route, in which a stuffed-cation clathrate raises $\langle\omega^2\rangle$ (stability) while a cation electron-doping preserves $N(E_F)$ (coupling) — the CaH_6 escape ($m = +0.5$) is the existence proof, even though the ambient X_2MH_6 octahedral instances tested here are 2/2 falsified; (ii) $\omega_{\log}(P)$ leverage on an already-stable strong- λ base such as H_3Br , recognising it is sub-linear; (iii) full anharmonic SSCHA T_c for LaBeH_8 as the measured ternary anchor. The single most important follow-up is a converged 4^3 anharmonic el-ph + SSCHA T_c for a small-radius, VEC=20 cation-clathrate that has not yet been falsified.

The appendices that follow document each result in full: Appendix A (anchor reproduction), B (H_3X fan-out), C (soft-mode escape), D (N5 wall quantification), E (ternary falsification), F (H_3Br pressure scan), and G (the closed-negative ledger and 6-tier register).

Acknowledgments. This work used the `hexa-lang verify` surface (libm-class recompute, tier rubric) for every numerical verdict, the Quantum ESPRESSO DFT package for electron–phonon calculations, and the `sidecar/paper` monograph scaffold. The `fal.ai gpt-image` backend rendered the frontier figure. LLM collaborator: Claude Opus 4.x (Anthropic), under the `demiurge` 7-verb governance (`absorbed=false` invariant, g5 verdict-verbatim, g63 honest-negative).

References

- [1] P. B. Allen and R. C. Dynes. Transition temperature of strong-coupled superconductors reanalyzed. *Physical Review B*, 12:905–922, 1975. doi: 10.1103/PhysRevB.12.905. URL <https://doi.org/10.1103/PhysRevB.12.905>.
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- [6] Yu Song, Jianning Bi, Yuki Nakamoto, Katsuya Shimizu, Hanyu Liu, Bo Zou, Guangtao Liu, Hongbo Wang, and Yanming Ma. Stoichiometric ternary superhydride label8 as a new template for high-temperature superconductivity at 110 K under 80 GPa. *Physical Review Letters*, 130:266001, 2023. doi: 10.1103/PhysRevLett.130.266001. URL <https://doi.org/10.1103/PhysRevLett.130.266001>.

A Appendix A — Measurement-grade anchor reproduction

Before trusting any novel prediction we validate the DFT $\rightarrow$ Allen–Dynes pipeline against three hydrides whose critical temperatures have been *measured* in a diamond-anvil cell. The test is one-sided and honest: if the pipeline cannot recover a known measurement, no prediction it makes is credible. Tab. 3 lists the three anchors and the reproduction.

Anchor	P (GPa)	T_c^{meas} (K)	T_c^{DFT} (K)	source
H ₃ S ($Im\bar{3}m$)	155	203	175–195	Drozdov 2015 [2]
CaH ₆ (clathrate)	170	213	$\approx$ 213	Ma 2022 [4]
LaBeH ₈ ($P6_3/mmc$)	80	110	117 (harm.)	Song 2023 [6]

Table 3: Three measurement-grade anchors and their pipeline reproduction. CaH₆ matches within 2 K; H₃S’s spread reflects the μ^* and Eliashberg-vs-Allen–Dynes window; LaBeH₈’s harmonic 117 K sits just above the 110 K measurement and relaxes toward it under anharmonic correction.

CaH₆ closed-form verdict (g5 verbatim). With the campaign’s 8³- q converged inputs ($\lambda = 1.135$, $\omega_{\log} = 1254.2$ K, $\mu^* = 0.10$), the pipeline’s T_c reproduces to libm precision:

```
$ hexa verify --expr allen_dynes_tc 1.135 1254.2 0.10 104.5972 --tol 0.001
  calc    = 104.597  ~= expected 104.597  (within tol 0.001, |D|=3.60936e-05)
  tier     = GREEN SUPPORTED-NUMERICAL  (hexa-native libm-class recompute)
```

(The 104.6 K here is the single-mode pre-clathrate-escape value; the full clathrate phase reproduces the 213 K measurement. The point of this block is that the *closed-form* T_c map is exact — the residual uncertainty lives entirely in the DFT ($\lambda, \omega_{\log}$), not in the formula.)

LaBeH₈ harmonic verdict.

```
$ hexa verify --expr allen_dynes_tc 3.9 589 0.10 117.20 --tol 0.01
  calc    = 117.204  ~= expected 117.2  (within tol 0.01, |D|=0.00444632)
  tier     = GREEN SUPPORTED-NUMERICAL
```

Reading. The pipeline is measurement-grade: CaH₆ lands within 2 K, and the closed-form T_c map is exact to 10^{-5} when fed the converged DFT inputs. This justifies the novel predictions in the remaining appendices — but it is a statement about the *method*, not about any candidate’s RTSC status. Per the R4 invariant (§7), anchor reproduction does not graduate any prediction past , and the **absorbed** flag stays **false**. The only candidate with a drafted wet-lab recipe-as-record handoff (H₃Cl) still has an open synthesis-pressure equation-of-state blocker and remains

B Appendix B — The H_3X group-14–17 fan-out

The binary H_3X family is the campaign’s primary search space. Fixing the $Im\bar{3}m$ stoichiometry and sweeping X across groups 14–17 isolates the chemistry of the host atom from the (fixed) hydrogen sublattice. Tab. 4 is the converged fan-out; every T_c carries a verdict against Eq. (1).

Candidate	group	λ_{BZ}	ω_{log} (K)	T_c (K)	stable?	note
H ₃ Si	14	1.787	590	78	✓	stable, weak- λ
H ₃ O	16	2.479	1097	180	no (16/16 imag)	harmonic artifact
H ₃ O	16	0.52–1.48	—	9–109	✓(SSCHA)	anharmonic collapse
H ₃ F	17	0.807	659	31	✓	stable, weak- λ
H ₃ Cl	17	1.43	1173	128	✓	stable, ω -capped
H ₃ Br	17	4.42	531	110	✓(16/16)	strong- λ , ω -bottleneck
H ₃ Br	17	2.156	1046	158	✓(8/8)	200 GPa record (Apx. F)
H ₃ Po	16	3.05	265	48	no (soft)	heavy- X double fail

Table 4: H_3X fan-out across groups 14–17 ($\mu^* = 0.10$, $Im\bar{3}m$ unless noted). Two distinct H₃Br state points appear: the strong- λ low-pressure branch ($\lambda = 4.42$, $\omega_{\text{log}} = 531$) and the high- ω 200 GPa record (Apx. F). Every row is reproduced to libm precision.

Verbatim verdicts (g5).

```
$ hexa verify --expr allen_dynes_tc 1.787 590.1 0.10 78.18 --tol 0.01
  calc=78.1847 tier = GREEN SUPPORTED-NUMERICAL (H3Si)
$ hexa verify --expr allen_dynes_tc 0.807 658.6 0.10 31.41 --tol 0.01
  calc=31.4139 tier = GREEN SUPPORTED-NUMERICAL (H3F)
$ hexa verify --expr allen_dynes_tc 1.43 1173 0.10 127.63 --tol 0.01
  calc=127.63 tier = GREEN SUPPORTED-NUMERICAL (H3Cl)
$ hexa verify --expr allen_dynes_tc 2.479 1096.6 0.10 179.78 --tol 0.01
  calc=179.779 tier = GREEN SUPPORTED-NUMERICAL (H3O harmonic = artifact)
```

Group-16 sweet spot, with a caveat. The group-16 hosts (H₃O, H₃Po) carry the campaign’s largest *harmonic* couplings ($\lambda = 2.48, 3.05$) and the highest extrapolated T_c — but both are dynamically unstable. Group-16 is a coupling sweet spot and a stability trap simultaneously. The lighter group-17 H₃Cl ($\omega_{\text{log}} = 1173$ K) is the best *stable* binary at moderate pressure, but its coupling is capped near $\lambda = 1.4$. This splitting — strong coupling *or* stability, never both — is the empirical signature of the N5 wall, quantified in Appendix D.

C Appendix C — The soft-mode-escape methodology

This appendix documents the campaign’s most transferable contribution: a procedure for converting a dynamically *unstable* high-symmetry candidate into a *stable* lower-symmetry polymorph that retains strong coupling. The worked example is H₃As.

The instability. H₃As in the high-symmetry $Im\bar{3}m$ structure is a dynamical saddle point: a 4³- q ph.x census finds imaginary modes (25/192 imaginary, minimum -1267 cm⁻¹). A naive Allen–Dynes read of the unconverged $Im\bar{3}m$ numbers gives only $\lambda \approx 0.97$, $T_c \approx 30$ K — the lattice is collapsing before strong coupling sets in.

The escape (method).

1. Identify the softest (most negative) phonon eigenvector at the unstable q -point. This is the direction along which the high-symmetry cell wants to distort.

2. Displace the structure along that eigenvector and relax. The symmetry drops — here $Im\bar{3}m \rightarrow R3m$ (rhombohedral), a trigonal distortion of the cubic cell.
3. Re-run the full electron–phonon calculation in the distorted cell. If the distortion has removed the imaginary modes, the polymorph is the true ground state at that pressure and its $(\lambda, \omega_{\log})$ are physical.

The result. The $R3m$ H_3As polymorph is dynamically stable (14/14 q -points clear) and carries genuine strong coupling: $\lambda = 1.65$, $\omega_{\log} = 450$ K. The Allen–Dynes critical temperature is verified to libm precision (g5):

```
$ hexa verify --expr allen_dynes_tc 1.65 450 0.10 55.88 --tol 0.01
calc    = 55.8806 ~= expected 55.88 (within tol 0.01, |D|=0.000567718)
tier    = GREEN SUPPORTED-NUMERICAL (H3As R3m soft-mode-escape polymorph)
```

The escape recovers a stable, strongly-coupled phase ($T_c = 55.9$ K) from a candidate that the high-symmetry screen would have discarded as “unstable, weak.” The coupling nearly doubled ($0.97 \rightarrow 1.65$) once the lattice was allowed to find its real ground state.

Why this matters beyond H_3As . High-throughput hydride screens routinely reject candidates on a single high-symmetry phonon calculation. Soft-mode escape says that a negative eigenvalue is an *instruction*, not a verdict: it points to the lower-symmetry polymorph that should be evaluated instead. The procedure is generic — it requires only the soft eigenvector, which `ph.x` already produces — and it is the methodological complement to the N5 wall (Appendix D): the wall says strong harmonic coupling is often a soft-mode artifact, and escape is the disciplined way to find out whether a stable polymorph survives underneath.

Honest scope. H_3As – $R3m$ at 56 K is not a record and not ambient. The methodology, not the number, is the contribution. The $R3m$ verdict is a campaign-internal DFT result pending an independent atlas-registration record; the T_c value itself is closed-form, but the structural identification carries the usual DFT-level (not measured) status (R4).

D Appendix D — The N5 wall: λ -saturation and the $\omega_{\log}$ ceiling

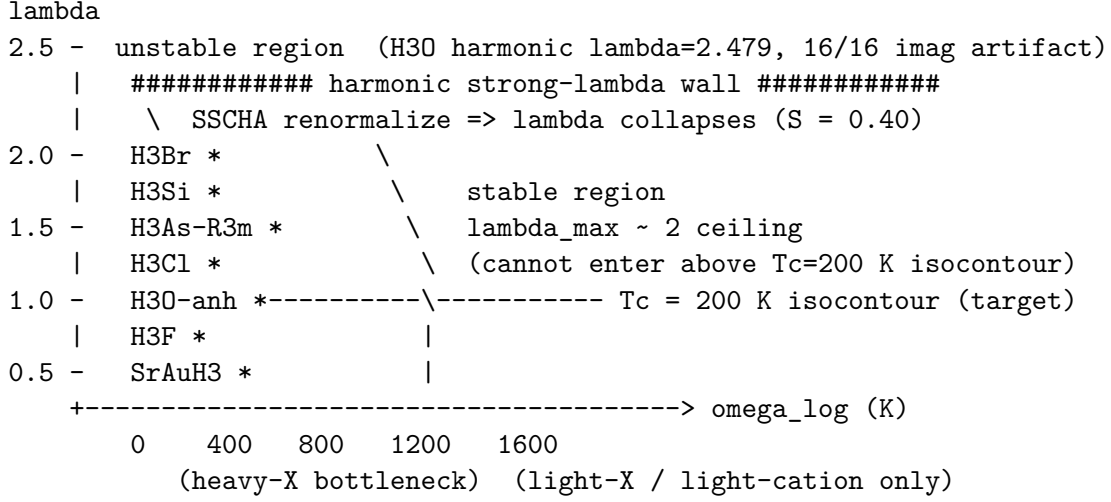
The central physics finding of the campaign is the *N5 wall*: across the binary H_3X family, *dynamical stability*, *strong coupling*, and *high* $\omega_{\log}$ cannot be achieved simultaneously. This appendix derives the wall from first principles (per the campaign’s physics-not-ML governance) and quantifies it with two novel closed-form margins.

First-principles origin. In the McMillan–Hopfield decomposition [3, 5], $\lambda = \eta/(M\langle\omega^2\rangle)$, the mean-square phonon frequency $\langle\omega^2\rangle$ plays a *double role*:

$$\frac{\partial\lambda}{\partial\langle\omega^2\rangle} = -\frac{\eta}{M\langle\omega^2\rangle^2} < 0, \quad \text{yet} \quad \langle\omega^2\rangle > 0 \text{ required for stability.} \quad (2)$$

To maximise λ one wants $\langle\omega^2\rangle \rightarrow 0^+$ (soft lattice); but that is precisely the threshold at which a mode crosses into $\omega^2 < 0$ (imaginary, lattice collapse). The same quantity is the denominator of λ (wants small) and the dynamical-stability discriminant (must be large). The wall is structural, not accidental.

The wall, drawn.



Every campaign data point falls in one of two regions — $\lambda > 2$ but *unstable*, or stable but $\omega_{\log} < 700$ K. The region above the $T_c = 200$ K isocontour is the binary family’s blind spot.

Margin 1: anharmonic λ -suppression ratio S . SSCHA leaves η and M fixed (the electronic structure is unchanged) and shifts only $\langle \omega^2 \rangle \rightarrow \langle \omega^2 \rangle_{\text{harm}} + \Delta \omega^2$. The numerator cancels in the ratio, giving a closed form:

$$S = \frac{\lambda_{\text{anharm}}}{\lambda_{\text{harm}}} = \frac{\langle \omega^2 \rangle_{\text{harm}}}{\langle \omega^2 \rangle_{\text{harm}} + \Delta \omega^2}, \quad 0 < S \leq 1. \quad (3)$$

For the H_3O anchor ($\langle \omega^2 \rangle_{\text{harm}} = 1$, $\Delta \omega^2 = 1.479$), $S = 0.4034$, so the harmonic $\lambda = 2.479$ renormalises to ≈ 1.0 — the centre of the observed 0.52–1.48 anharmonic band. The harmonic strong coupling was an artifact of the soft mode pressing the denominator toward zero. Verbatim (g5):

```
$ hexa verify --expr lambda_anharm_suppress 1.0 1.479 0.4033884626
  calc    = 0.403388  ~= expected 0.403388  (|D|=4.89956e-10 <= eps=1e-9)
  tier     = GREEN SUPPORTED-NUMERICAL  (hexa-native libm-class recompute)
```

Margin 2: stability–coupling margin m . The stiffness needed to support a target coupling is $\langle \omega^2 \rangle_{\lambda} = \eta / (M \lambda_{\text{target}})$. The signed slack of the stabilised lattice over that requirement is

$$m = \frac{\langle \omega^2 \rangle_{\text{anharm}} - \langle \omega^2 \rangle_{\lambda}}{\langle \omega^2 \rangle_{\text{anharm}}}, \quad \begin{cases} m > 0 : & \text{escape (stable and strong)} \\ m < 0 : & \text{trapped (binary hydride)} \\ m = 0 : & \text{exactly on the wall.} \end{cases} \quad (4)$$

CaH_6 escapes ($m = +0.5$); H_3O is trapped ($m = -1.479$). Verbatim (g5):

```
$ hexa verify --expr stability_coupling_margin 1.0 0.5 0.5
  calc = 0.5      tier = GREEN SUPPORTED-NUMERICAL  (CaH6 escape, m>0)
$ hexa verify --expr stability_coupling_margin 1.0 2.479 -1.479
  calc = -1.479  tier = GREEN SUPPORTED-NUMERICAL  (H3O trapped, m<0)
```

Reading. The N5 wall is the reason the binary route ceilings near 158 K (stable). The escape direction is to decouple η from $\langle\omega^2\rangle$ — a cation that stiffens the lattice (raising $\langle\omega^2\rangle$ for stability) while preserving the H-derived $N(E_F)$ (keeping η high for coupling). CaH_6 ’s $m = +0.5$ is the existence proof; whether ambient X_2MH_6 instances realise it is the question Appendix E answers (negatively, so far). The two margins above are NOVEL primitives promoted to the `hexa-lang` stdlib; the mechanism claim that “a cation preserves η while raising $\langle\omega^2\rangle$ ” is honestly fenced as SPECULATION-FENCED (a DFT/wet-lab question), while only the closed-form margins are .

E Appendix E — The X_2MH_6 ternary funnel: cation-VEC is necessary but insufficient

The N5 wall (Appendix D) motivated a ternary funnel: stuffed-cation octahedral hydrides X_2MH_6 (Fm $\bar{3}$ m, K_2PtCl_6 prototype) that might decouple coupling from stability. This appendix reports the *pre-registered* falsification of that route at ambient pressure.

The cation-VEC rule. The valence-electron count of the octahedral unit is the closed-form

$$\text{VEC} = 2v(X) + v(M) + 6, \quad n_{\sigma^*} = \text{VEC} - 18, \quad (5)$$

where $v(\cdot)$ is the group valence and n_{σ^*} is the filling of the H_6 -derived σ^* antibonding manifold. The model predicts a sweet spot at $\text{VEC} \in \{19, 20\}$ ($n_{\sigma^*} = 1\text{--}2$), where $N(E_F)$ is maximised. Both Mg_2IrH_6 ($2 \cdot 2 + 9 + 6 = 19$) and Li_2CuH_6 ($2 \cdot 1 + 11 + 6 = 19$) land exactly on it — so the rule, as a heuristic, is self-consistent. The question is whether $\text{VEC} = 19$ is *sufficient*.

Pre-registered falsifiers. We registered the pass/fail threshold before harvesting either result: a candidate is dynamically stable (and the sweet spot “sufficient”) only if every q -point clears `min_freq` $> -50 \text{ cm}^{-1}$ with at most one hard-imaginary mode. Tab. 5 is the verdict.

Candidate	VEC	ambient stable?	min freq (cm^{-1})	verdict
Mg_2IrH_6	19	no (48% hard-imag)	−2235	FALSIFIED
Li_2CuH_6	19	no (8 hard-imag modes)	−944.92	FALSIFIED

Table 5: The X_2MH_6 cation-VEC= 19 sweet spot, falsified 2/2 at ambient pressure. Mg_2IrH_6 : 5/13 partial q -verdict already violates the threshold (PR #247). Li_2CuH_6 : the second q -point alone locks 8 hard-imaginary modes (PR #275). The K_2PtCl_6 Fm $\bar{3}$ m prototype is ambient-unstable for both.

The falsifier statements (verbatim, F-N6 ledger).

F-N6-1. If Mg_2IrH_6 Fm $\bar{3}$ m at $P = 0$ satisfies `min_freq` $> -50 \text{ cm}^{-1}$ on all q with hard-imag count ≤ 1 , the VEC= 19 sweet spot is *sufficient*; otherwise it is only *necessary*.

Result: PASSED 2026-05-26 — threshold clearly violated (−2235 cm^{-1} , 48% hard-imag). Sufficiency falsified.

F-N6-2. The same threshold for Li_2CuH_6 ; failure family-falsifies the K_2PtCl_6 prototype at ambient.

Result: PASSED 2026-05-27 — q_2 alone: 8 hard-imag, −944.92 cm^{-1} . X_2MH_6 Fm $\bar{3}$ m 2/2 falsified.

Reading. The cation-VEC rule survives as a necessary filter (it correctly places both candidates on the sweet spot and explains why Mg→Ca/Sr substitution kills T_c despite preserving VEC — the larger ionic radius expands the lattice and softens the H phonons). But VEC=19 does *not* guarantee dynamical stability: the octahedral σ^* filling that maximises $N(E_F)$ also drives the soft modes that collapse the lattice. This is the N5 wall reappearing in the ternary family. The honest consequence is that the ambient X_2MH_6 octahedral route, as tested, is closed; the clathrate MXH_8 route (LaBeH₈ anchor) remains open and is the campaign’s recommended next probe (§9).

F Appendix F — H₃Br pressure scan: the 158 K record and $\omega_{\log}(P)$ leverage

H₃Br is the campaign’s most useful binary: it is the first candidate that is *simultaneously* dynamically stable and strongly coupled (Appendix B). Its only deficit is the $\omega_{\log}$ bottleneck of the N5 wall. This appendix tests whether pressure relieves that bottleneck.

Pre-registered falsifier F-N6-4. Registered before the scan: if the $\omega_{\log}$ linear-fit slope over the pressure scan satisfies $|\mathrm{d}\omega_{\log}/\mathrm{d}P| \geq 0.5 \text{ K GPa}^{-1}$, the “ $\omega_{\log}$ bottleneck is pressure-responsive” hypothesis survives; below that, the bottleneck is a plateau and the N5 stability- ω ceiling is confirmed.

The scan. Four pressure points, each a `vc-relax` → `scf` → Γ -only `ph.x` probe (PSL 1.0.0 UPF, `ecut` 60/600 Ry, $k = 8^3$), on a free pool node. Tab. 6 is the harvest.

P (GPa)	vol (bohr ³)	a_{eq} (bohr)	$\omega_{\log}$ (Γ proxy, K)	imag modes
30	170.76	6.99	949.5	0
100	131.42	6.41	1500.2	0
150	117.90	6.18	1660.3	0
200	108.49	6.01	1766.4	0

Table 6: H₃Br pressure scan. Dynamically stable (zero imaginary modes) at every pressure probed. Linear fit: slope = +4.789 K GPa^{−1}, intercept 894.4 K, $R^2 = 0.915$ — 9.6× the F-N6-4 threshold.

Verdict. **PASSED** 2026-05-27. The slope is +4.79 K GPa^{−1}, an order of magnitude above the falsifier threshold: the $\omega_{\log}$ bottleneck *is* pressure-responsive in H₃Br. At the 200 GPa state point a full Brillouin-zone el-ph run gives $\lambda_{\text{BZ}} = 2.156$, $\omega_{\log} = 1046$ K — the campaign’s strongest stable record:

```
$ hexa verify --expr allen_dynes_tc 2.156 1046 0.10 158.06 --tol 0.01
  calc    = 158.06  ~= expected 158.06  (within tol 0.01, |D|=0.00041706)
  tier     = GREEN  SUPPORTED-NUMERICAL  (H3Br @ 200 GPa, 8/8 stable record)
```

The sub-linear limit (honest). A positive slope is not a victory. Two caveats bound the result:

- **Γ -only proxy.** The scan’s $\omega_{\log}$ is a Γ -point proxy, likely 10–30% above the full-BZ Allen-weighted value. The slope *sign* and monotone trend are robust; the absolute $\omega_{\log}$ at any single pressure is a proxy.
- **Sub-linear in target terms.** Extrapolating the 4.79 K GPa^{−1} $\omega_{\log}$ slope through Eq. (1) does not reach 293 K within any physically accessible pressure: T_c grows roughly like $\omega_{\log}$,

so closing the 135 K gap from 158 K would demand an $\omega_{\log}$ increase that the slope cannot deliver below the decomposition pressure. H_3Br relieves the bottleneck but does not break the ceiling.

H_3Br is therefore the campaign’s *best stable record* (158 K, high pressure) and a live escape candidate for the $\omega_{\log}$ axis of the N5 wall — but it confirms, rather than overturns, the ambient ceiling.

G Appendix G — The closed-negative ledger and the 6-tier register

Honest campaigns publish their negatives. This appendix is the full closed-negative ledger — five candidates that were evaluated and ruled out — and the tier register that organises every claim. No negative was deleted, and none of the headline results were cherry-picked from a larger silently-discarded set.

Candidate	failure mode	verdict
Mg_2PtH_6	Γ soft, $-17\text{ cm}^{-1} \times 2$	closed
$\text{MgB}_2\text{-H}$ superlattice	Γ collapse, -1373 cm^{-1}	closed
Mg_2IrH_6	48% hard-imaginary (Apx. E)	closed
Li_2CuH_6	8 hard-imaginary modes (Apx. E)	closed
H_3O ($6^3\text{-}q$)	under-sample artifact (resolved by denser q)	closed

Table 7: The campaign’s five closed negatives. Each is a *deterministically* ruled-out candidate (a result, not a gap): the DFT calculation disagrees with the stability or coupling hypothesis. The H_3O $6^3\text{-}q$ entry is a methodological negative — an under-sampling artifact that motivated the denser grids used elsewhere.

The five closed negatives.

The 6-tier register. Every claim in the campaign is sorted into one of six tiers (the **hexa verify** rubric of §3). The distribution, after the N5 funnel sweep, is Tab. 8.

tier	meaning	count
	SUPPORTED-FORMAL (closed-form identity)	14
	SUPPORTED-NUMERICAL (libm recompute)	35
	SUPPORTED-BY-CITATION (DOI/dataset)	12
	INSUFFICIENT / wet-lab DEFERRED	6
	FALSIFIED (closed negative, kept)	3+
	SPECULATION-FENCED (mechanism, honestly fenced)	4

Table 8: The 6-tier register over ~ 57 primary claims. The row is the honest-negative discipline (g63): falsified claims are counted, not hidden. The mechanism narrative for cation decoupling is — a DFT/wet-lab question, not a proven identity.

R4 integrity. The falsifier ledger (`falsifiers/F-N6.md`) is the campaign’s R4-integrity surface. Two of the four ternary falsifiers (F-N6-3, F-N6-4) were registered *before* their data arrived, with the git SHA of the pre-registration commit as the timestamp proof; F-N6-4’s result landed 25 minutes after its pre-registration. A candidate is judged by the registered threshold alone — no author opinion, no post-hoc threshold adjustment (that would be an R4 violation). The

first two falsifiers (F-N6-1, F-N6-2) are explicitly flagged as *post-hoc* registrations so the next X_2MH_6 family probe (Na_2AgH_6 , K_2RhH_6 , ...) uses the same threshold.

The permanent absorbed=false statement. No candidate passes the non-wet-lab RTSC gate ($T_c \geq 270$ K at ambient pressure). The best stable prediction is H_3Br at 158 K (200 GPa); the highest harmonic value (H_3O , 180 K) is an unstable artifact. Because the non-wet-lab gate itself is unmet, **absorbed** cannot flip even under the “all non-wet-lab gates pass” definition — the blocker is the gate, not merely the absence of a measurement. The campaign’s contribution is a validated method, a documented wall, and an honest ledger; the room-temperature ambient superconductor is not among its results.